

Stochastic Process: Problems

Dr. Harshitha A.

1 $P(X_{m+1} = i | X_m = j) = p_{ji}$

2

$$\begin{aligned} P(X_0 = i_0, X_1 = i_1, X_2 = i_2, \dots, X_n = i_n) &= P(X_0 = i_0) \times P(X_1 = i_1 | X_0 = i_0) \\ &\quad \times P(X_2 = i_2 | X_1 = i_1) \times \dots \\ &\quad \times P(X_n = i_n | X_{n-1} = i_{n-1}) \\ &= P(X_0 = i_0) \times p_{i_0 i_1} \times p_{i_1 i_2} \times \\ &\quad \dots \times p_{i_{n-1} i_n} \end{aligned}$$

3

$$\begin{aligned} &P(X_0 = i_0, X_1 = i_1, X_2 = i_2, \dots, X_n = i_n | X_0 = i_0) = \\ &\frac{P(X_0 = i_0)P(X_1 = i_1 | X_0 = i_0)P(X_2 = i_2 | X_1 = i_1) \dots P(X_n = i_n | X_{n-1} = i_{n-1})}{P(X_0 = i_0)} \\ &= \frac{P(X_0 = i_0) \times p_{i_0 i_1} \times p_{i_1 i_2} \times \dots \times p_{i_{n-1} i_n}}{P(X_0 = i_0)} \end{aligned}$$

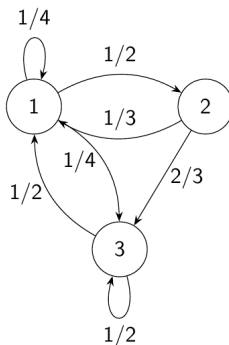
Example

1. An urn contains 5 white balls and 5 black balls. The following experiment is repeated indefinitely: A ball is drawn from the urn. If the ball is white, it is put back in the urn. Otherwise, it is left out. Let X_n be the number of black balls remaining in the urn after n draws from the urn. Find the TPM.

Example

2. An urn contains 3 white balls and 2 black balls. The following experiment is repeated indefinitely: A ball is drawn from the urn. If the ball is white, it is put back in the urn. Otherwise, it is left out. Let X_n be the number of black balls remaining in the urn after n draws from the urn. Find the TPM.

3. The transition diagram is,



- i Find $P(X_4 = 3 | X_3 = 2)$
- ii Find $P(X_3 = 1 | X_2 = 1)$
- iii If we know $P(X_0 = 1) = \frac{1}{3}$, find $P(X_0 = 1, X_1 = 2)$
- iv If we know $P(X_0 = 1) = \frac{1}{3}$, find $P(X_0 = 1, X_1 = 2, X_2 = 3)$
- v Find Transition probability matrix.

4. Let $\{X_n, n = 0, 1, 2, \dots\}$ be a Markov chain with state space $\{0, 1, 2\}$ and the initial probability distribution $P(X_0 = i) = \frac{1}{3}, i = 0, 1, 2$. One step TPM is given by

	0	1	2
0	$\frac{3}{4}$	$\frac{1}{4}$	0
1	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$
2	0	$\frac{3}{4}$	$\frac{1}{4}$

- i Find $P(X_1 = 1 | X_0 = 2)$
- ii Find $P(X_2 = 2 | X_1 = 1)$
- iii Find $P(X_2 = 2, X_1 = 1 | X_0 = 2)$
- iv Find $P(X_2 = 2, X_1 = 1, X_0 = 2)$
- v Find $P(X_3 = 1, X_2 = 2, X_1 = 1, X_0 = 2)$

Higher transition probabilities

- One-step probability from i to j is

$$P_{ij}^{(1)} = P(X_{n+1} = j | X_n = i)$$

- Two-step probability from i to j is

$$P_{ij}^{(2)} = P(X_{n+2} = j | X_n = i)$$

- m -step probability from i to j is

$$P_{ij}^{(m)} = P(X_{n+m} = j | X_n = i)$$



$$P^{(1)} = [P_{ij}]$$

$$P^{(2)} = P^{(1)} P^{(1)}$$

$$P^{(m+1)} = P^{(m)} P^{(1)}$$

$$P^{(m+n)} = P^{(m)} P^{(n)}$$

Definition (Probability vector)

It is a vector $V = \{v_1, v_2, \dots, v_n\}$ such that $v_i \geq 0$ for all i and $\sum_{i=1}^n v_i = 1$

Definition (Stochastic matrix)

It is a square matrix with each row being a probability vector.

Definition (Fixed vector)

A vector v is said to be a fixed vector or a fixed point of a matrix A if $vA = v$ and $v \neq 0$.

Definition (Probability distribution)

Probability distribution of the system n – steps later is given by

$$P^{(n)} = P^{(0)} P^n$$

i.e.,

$$P^{(1)} = P^{(0)} P$$

$$P^{(2)} = P^{(0)} P^2 = P^{(1)} P$$

$$P^{(3)} = P^{(0)} P^3 = P^{(2)} P$$

and so on.

Theorem

Let P be a regular transition matrix of a Markov chain. Then in the long run, the probability that any state a_j occurs is approximately equal to the component t_j of the unique fixed probability vector t of P .

Example

1. Find the unique fixed probability vector t of

$$P = \begin{bmatrix} 0 & \frac{3}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Example

2. Every year, a man trades his car for a new car. If he has a Maruti, he trades it for an Ambassador. If he has an Ambassador, he trades it for a Santro. However, if he has a Santro, he is just as likely to trade it for a new Santro as to trade it for a Maruti or an Ambassador. In 2000, he bought his first car which was a Santro.

- (i) Find probability that he has
 - (a) 2002 Santro
 - (b) 2002 Maruti
 - (c) 2003 Ambassador
 - (d) 2003 Santro
- (ii) In the long run, how often will he have a Santro?

Example

3. For a Markov chain, the transition matrix is

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{3}{4} & \frac{1}{4} \end{bmatrix}$$

with initial distribution $P^{(0)} = (\frac{1}{4} \ \frac{3}{4})$. Find

- 1 $P_{21}^{(2)}$
- 2 $P_{12}^{(2)}$
- 3 $P^{(2)}$
- 4 the vector $P^{(0)}P^n$ approaches.
- 5 P^n approaches.

Example

4. A salesman's territory consists of 3 cities A, B and C. He never sells in the same city on successive days. If he sells in city A, then the next day he sells in city B. However, if he sells in either B or C, then the next day he is twice as likely to sell in city A as in other city. In the long run, how often does he sell in each of the cities?

Example

5. In a certain market, only two brands of cold drinks A and B are sold. Given that a man last purchased brand A, there is 80% chance that he would buy same brand in the next purchase. While if the man purchased brand B, there is 90% chance that his next purchase would be brand B. Using this information,

- 1 Develop TPM
- 2 Determine the long-run loyalty for brand A and brand B.